

**MAE-Access
V100R021C10**

Open API Developer Guide (Wireless Network)

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1 About This Document

Purpose

An open API is a REST-based open interface provided by the MAE NBI system. REST is short for Representational State Transfer. Third-party developers can use open APIs to access open MAE resources, including alarm, performance, and MML modules.

Product Version

Product Name	Product Version
MAE	V100R021C10

2 Open APIs

- [2.1 Introduction to Open APIs](#)
- [2.2 Introduction to REST](#)
- [2.3 Introduction to JSON](#)
- [2.4 API Communication Protocols](#)

2.1 Introduction to Open APIs

Open APIs use open authentication. An asymmetric encryption technology is used to authenticate users, obtain resources, and implement data sharing among platforms, enterprises, and entities. Open APIs are common applications at service-oriented websites. A website service provider encapsulates its website services in a series of APIs and enables the APIs for third-party developers. This is called APIs for opening websites (open APIs for short).

Open APIs provide a standard integration solution that facilitates quick integration of third-party systems. Open APIs can be used only through HTTPS upon authorization. This increases security. Data is exchanged in JavaScript object notation (JSON) format to simplify data read and write as well as to reduce the volume of consumed traffic in comparison with the XML format.

2.2 Introduction to REST

Representational State Transfer (REST) is a design and development method for network applications. It decreases development complexity and improves system scalability. The design concepts and principles of this architecture style are as follows:

1. All entities on a network can be abstracted as resources.
2. Each resource has a unique resource identifier, and operations on resources do not change the resource identifiers.
3. All operations are stateless.
4. Four standard methods are used to operate resources: POST, GET, PUT, and DELETE.
 - POST is used to create a resource.
 - GET is used to query resource information.

- PUT is used to update a resource.
- DELETE is used to delete a resource.

2.3 Introduction to JSON

JSON is a lightweight, standard data format. JSON strings are more concise than XML, thus occupying less space and decreasing network traffic. JSON has the following two structures:

- Object: Braces ({}) hold objects, for example, {key1: value1, key2: value2, ...}. You can access data using Object.key.
- Array: Brackets ([]) hold arrays, for example, [element1, element2, ...]. You can access data using Object[index]. The index starts from 0.

The following code shows how to obtain JSON strings.

```
{
  "id": "00000002",
  "href": "/api/rest/configurationManagement/v1/bulkCM/exportJobs/00000002",
  "userName": "testuser",
  "exportPath": "/export/home/omc/var/fileint/cm/0/",
  "reportPath": "/export/home/omc/var/fileint/cm/0/report",
  "errorInfo": ""
}
```

2.4 API Communication Protocols

All APIs for the NBI system are HTTPS-based RESTful APIs.

HyperText Transfer Protocol (HTTP) is one of the most widely used network protocols in Internet.

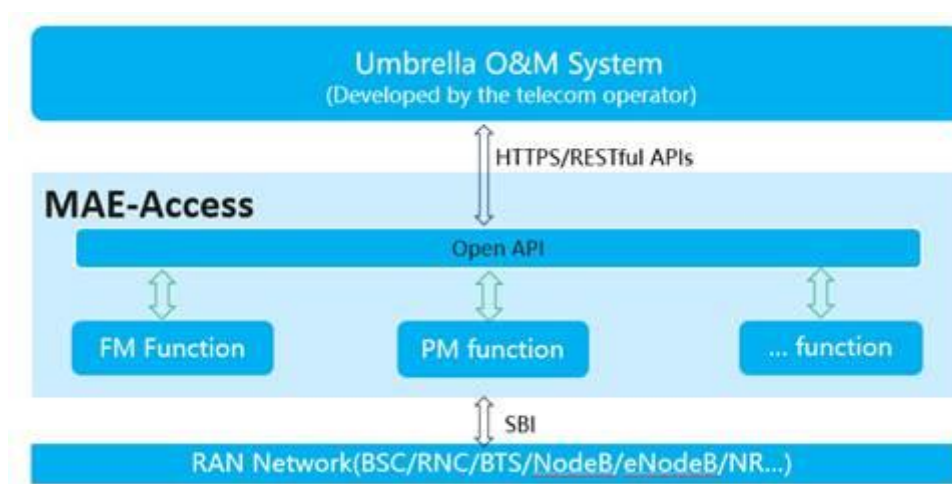
HTTP is designed to provide a method of releasing and receiving HTML pages.

Resources requested according to HTTP or Hypertext Transfer Protocol over Secure Sockets Layer (HTTPS) are identified by their Uniform Resource Identifiers (URIs).

Hypertext Transfer Protocol over Secure Socket Layer (HTTPS) is a secure HTTP channel. HTTPS is designed based on SSL. Therefore, data is encrypted using SSL to ensure data transmission security.

3 API Architecture and Functions

Each service releases some interfaces through northbound open APIs for third-party systems to call.



4 API Interconnection Process

- [4.1 Preparing for the Interconnection](#)
- [4.2 Creating a Third-Party User on the OSS](#)
- [4.3 \(Optional\) Deploying iSStar Scripts](#)
- [4.4 Verifying the Interconnection](#)

4.1 Preparing for the Interconnection

You need to obtain the following information before the interconnection.

Parameter	Value	Description
IP address of the application plane in the MAE system	Determined based on site conditions	Used for API access.
Port on the API gateway in the MAE system	31127	Used for API access.

4.2 Creating a Third-Party User on the OSS

Creating a User

Before calling APIs, the OSS needs to create users and bind roles to the users.

For information related to user creation, see the user management sections in the product documentation. The process of creating a user is as follows.

Step 1 Log in to MAE.

Open a web browser, enter **https://MAE IP address:31943/** (for example, **https://*.*.*.31943/**) in the address box, and press **Enter**.

Step 2 Creating a User

1. On the main menu, choose **Common > Security > User Management > Users**. On the displayed page, click **Create**.
2. Set **Type** to **Third-party**, set user information, and click **Next**.
3. In the role list, select **NBI User Group** and click **Next**.
4. Click **OK**.

Step 3 (Optional) If the MML interface needs to be called, grant the MML permission to the user.

In the **Apps** area of the MAE home page, click **Access** to enter the MAE-Access home page. On the main menu, choose **Security > NE User Management**. In the navigation pane, choose **MML Command Group Manager**. Select the user created in [Step 2](#) based on the user type, grant MML command groups to the user based on the NE types contained in the managed objects, and click **Apply**.

----End

4.3 (Optional) Deploying iSStar Scripts

Step 1 Develop iSStar script applications.

1. Scripts are in .hsp3 format. You can choose **MAE > Access > Maintenance > iSStar > Development Platform** to develop .hsp3 scripts or choose **Application Platform > Application Management** to download the developed applications.
2. To ensure security and reliability, do not access scripts outside the .hsp3 package in the .hsp3 script. For example, do not use the Call, Import, and CreateTask functions to invoke scripts outside the .hsp3 package.
3. Operations on multiple NEs in a single task are executed in sequence. Therefore, you are not advised to perform time-consuming operations on each NE when a large number of NEs are involved.

Step 2 Use SFTP or FTPS to upload the script application to the MAE master service node.

Use FileZilla to upload the **test.hsp3** application to the **/home/sopuser** directory on the MAE master service node as the **sopuser** user.

Step 3 Create a directory on the MAE server.

1. Create a script repository.
Use PuTTY to log in to the MAE master service node as the **sopuser** user in SSH mode.
Create a folder in the **/export/home/sysm/ftproot/** directory for storing scripts.

```
$ su ossuser  
$ cd /export/home/sysm/ftproot/  
$ mkdir iSStarScriptStore  
$ chmod 770 iSStarScriptStore
```
2. Create a task script directory.
Create the required directory in the **iSStarScriptStore** directory based on service scenarios and task types. The RESTful interface for creating tasks needs to transfer the directory name as a parameter to the RESTful server.
For example, a task directory can be created by running the following commands:

```
$ cd /export/home/sysm/ftproot/iSStarScriptStore  
$ mkdir demo
```

```
$ chmod 770 demo
$ cp /home/sopuser/test.hsp3 /export/home/sysm/ftproot/iSStarScriptStore/demo/
$ cd demo
$ chmod 770 test.hsp3
```

Step 4 Check the files that have been uploaded to the MAE server.

Use PuTTY to log in to the MAE master service node as the **sopuser** user in SSH mode.

Run the following commands to obtain the sha256 verification value of the file.

```
$ su ossuser
$ cd /export/home/sysm/ftproot/iSStarScriptStore/demo
$ sha256sum /export/home/sysm/ftproot/iSStarScriptStore/demo/test.hsp3
----End
```

NOTE

1. To call iSStar APIs, you need to deploy iSStar scripts.
2. In this document, *demo* and *test.hsp3* are used as examples. Replace them as required.

4.4 Verifying the Interconnection

Debugging APIs Using Postman

This section uses the Chrome web browser as an example to describe how to use Postman to debug an API.

Prerequisites

You have obtained the required IP address and port number. For details, see 4.1 Preparing for the Interconnection.

You have created a northbound login authentication user. For details, see 4.2 Creating a Third-Party User on the OSS.

Obtaining the Token

Step 1 Start Postman, and enter the following URL in the address box:

https://Primary IP address of the application plane in the MAE system:31127/api/rest/securityManagement/v1/oauth/token

Step 2 Set the request header parameters.

Content-Type: application/json

Accept: application/json

Step 3 Set the request body parameters.

The request type is PUT. The body parameters are as follows:

```
{
```

```
"grantType": "password",  
"userName": "",  
"value": ""  
}
```

 NOTE

1. You need to switch the request body parameters to the raw mode.
1. You need to replace userName and value with the user name and password of the user for NBI system login authentication.

Step 4 Send the URI. If status code 200 is returned, the request is successful.

Step 5 View the response.

```
{  
  "accessSession":  
    "x-c97tk99iarc6kbo5hhjz09bw2n6rdeo76r08ukjw9gvxpders4ikjt5iqo0a05tgk805lfpcpeb  
    wlf7vbvthcuoimle6obz49lcli4bbspc2len9evsalk8sant9hhc",  
  "roaRand": "5ac18de46d9a7b72d3ea5ad7877ff6d566a2a24b2078f48e",  
  "expires": 1800,  
  "additionalInfo": null  
}
```

 NOTE

accessSession indicates the returned token.

When calling an API, you need to fill the value of accessSession in the X-Auth-Token field.

The following is an example:

X-Auth-Token:x-c97tk99iarc6kbo5hhjz09bw2n6rdeo76r08ukjw9gvxpders4ikjt5iqo0a05tgk805lfpcpebwlf7vbvthcuoimle6obz49lcli4bbspc2len9evsalk8sant9hhc

----End

Calling a Northbound API (Example)

Step 1 In the Postman app, enter an URI in POST mode.

https://Primary IP address of the application plane in the MAE system:31127/api/rest/performanceManagement/v1/measurementResults

Step 2 Set the request header parameters.

Content-Type:application/json

X-Auth-Token:x-c97tk99iarc6kbo5hhjz09bw2n6rdeo76r08ukjw9gvxpders4ikjt5iqo0a05tgk805lfpcpebwlf7vbvthcuoimle6obz49lcli4bbspc2len9evsalk8sant9hhc

 NOTE

The value of X-Auth-Token is the value of accessSession obtained in "Obtaining the Token."

Step 3 Set the request body parameters.

```
{
```

```
"**", "*****"
```

```
}
```

Step 4 Send the URI. If status code 200 is returned, the API is successfully called.

```
{
```

```
"**", "*****"
```

```
}
```

----**End**

5 API List

- [5.1 List of Public APIs](#)
- [5.2 List of MML APIs](#)
- [5.3 List of Alarm APIs for FM](#)
- [5.4 List of Performance APIs for PM](#)
- [5.5 API for Querying Topology Cell Information](#)
- [5.6 List of iSStar APIs](#)
- [5.7 List of System Information APIs](#)

5.1 List of Public APIs

5.1.1 Login API

Function

This API is used for logging in to MAE using the name and password of a user account. A session ID that uniquely identifies the login session is returned after a successful login.

Precautions

- If the logged-in user has not performed any operations within 30 minutes, the session ID automatically becomes invalid by default.
- You need to create users and bind roles to the user for interconnecting with third-party systems.
- If the password is entered incorrectly for five consecutive times, the user account will be locked. In this case, you need to manually unlock the user account on the OSS.

Call Method

PUT

URI

/api/rest/securityManagement/v1/oauth/token

Request Parameters

Table 5-1 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
Content-Type	Request header	STRING	Type of the request message body. The value of this parameter is application/json .	Mandatory

Table 5-2 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
TokenRequest	Yes	REFERENCE	For details, see Table 5-3.		It is used for login authentication.

Table 5-3 Parameters in the TokenRequest MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
grantType	Yes	STRING	-	password	Authentication type.
userName	Yes	STRING	-	-	User name for logging in to MAE.
value	Yes	STRING	-	-	Password for logging in to MAE.

Sample Request

```
PUT
/api/rest/securityManagement/v1/oauth/token
Host: 10.10.10.10:31127
Content-Type: application/json
Accept: application/json
Accept-Language: en-US
{
  "grantType": "password",
  "userName": "test",
  "value": "Changeme 123"
}
```

Response Parameters

If the returned status code is **200**, the login is successful.

Table 5-4 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
TokenResponse	No	REFERENCE	For details, see Table 5-5.		It is used for login authentication.

Table 5-5 Parameters in the TokenResponse MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
accessSession	Yes	STRING	-	-	Session ID returned after a successful login. To call another API, you need to set the value of this parameter in the X-Auth-Token field in the request message header.
roaRand	Yes	STRING	-	-	Token returned after a successful login
expires	Yes	STRING	-	-	Expiration time of a session ID, in seconds.
additionalInfo	No	OBJECT	-	-	Reserved parameter.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Date: Wed, 21 Nov 2018 10:00:00 GMT
Server: example-server
Content-Type: application/json
{
  "accessSession":
"x-o5gb05o8eolfapgb49lj7wfsugbytgvxry7w0ahds9lic59dpdqnnuga9f9gnuldg8mko7bxtigaumarlfeltis8arhjioqlk8lecb891i5ctcbtemjxnt3x6nqr8asb",
  "roaRand": "58a68cd75382dd55189d0b855463fc2370b994bfdbfdead27",
  "expires": 1800,
  "additionalInfo": null
}
```


Operation Severity

Minor

5.2 List of MML APIs

5.2.1 Issuing a Single MML Command

Function

The function allows you to issue an MML command to the specified NEs. You can specify multiple NE names at a time.

Precautions

- Each specified NE name must be unique.
- If the number of specified NE names exceeds 100, you are advised to use a script. The timeout period for requesting the bus is 5 minutes. If there are too many NEs, the request may time out.
- The single-command issuance API displays only messages returned in synchronous mode. For asynchronous commands such as data synchronization, download, and version activation commands, the single-command issuance API does not return progress and result messages reported by NEs in asynchronous mode. If such query mode is used, you can issue the script to the NE.
- The maximum size of the request body is 2 MB.
- The maximum size of a response message is 10 MB.
- A maximum of 15 concurrent requests are supported.
- For the MML commands that need to run for a long time or whose response messages are large, you are advised to run them in MML batch script task issuing mode.
- If multiple NE names that are specified include incorrect name, an error is reported.

Call Method

POST

URI

/api/rest/mmlManagement/v1/command

Request Parameters

Table 5-6 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	STRING	Login session ID. The value of this parameter is the same as that of the	Mandatory

Parameter	Location	Type	Description	Mandatory/Optional
			accessSession parameter in the response message body for the API used for obtaining the token.	
Content-Type	Request header	STRING	Type of the request message body. The value of this parameter is application/json .	Mandatory

Table 5-7 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
Command Info	Yes	REFERENCE	For details, see Table 5-8.		Command information that is issued, including the command and specific NE information.

Table 5-8 Parameters in the CommandInfo MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
command	Yes	STRING	-	-	Specific MML command, for example, LST VER;;
neNames	Yes	ARRAY OF STRING	-	-	Name list of NEs where the command is executed

Sample Request

```

POST /api/rest/mmlManagement/v1/command
Host: serverIP:port
Content-Type: application/json; charset=UTF-8
X-Auth-Token:
x-jupj3v47dfhf4bpjnyermo2k9jpftdmmfveqgbmpf2kjpgaikfwo5rskbjulisbbwk89ghcqovwruar
ftlefvanfy5ctes9hehi2mfyoa5grzmk9fldhcft07g52rka
Content-Length:...
{
  "command": "LST VER;;",
  "neNames": ["pml", "pTest"]
}

```

Response Parameters

If the returned status code is **200**, the operation is successful.

Table 5-9 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
asynchId	No	STRING	-	-	Unique ID of the OSS.
results	No	ARRAY OF Result	For details, see Table 5-10.		MML result information. Each NE has a result.
retCode	No	STRING	-	-	Return code.
retMessage	No	STRING	-	-	Returned message.

Table 5-10 Parameters in the Result MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
name	Yes	STRING	-	-	Name of an NE.
report	Yes	STRING	-	-	NE response message. For details about the message format, see 5.2.1.1 Format of MML NE Response Messages.
result	Yes	STRING	-	-	Operation result.
retCode	No	INTEGER	-	-	Return code of MML command output. 0 indicates success. Others indicate failure.
serialId	No	INTEGER	-	-	Unique ID of a subsequent message received from the NE. The value -1 indicates that no subsequent message is received.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=UTF-8
Content-Length:...
{
  "asyncId": "0301 20210302193828 303",
  "results": [{
    "name": "pml",
    "report": "+++   LTE           2021-03-02 19:38:47\r\nO&M
#2440\r\n%/*1879048219*/LST VER:;%\r\nRETCODE = 0  Operation
succeeded.\r\n\r\nResult of current software
query\r\n-----\r\n      Current Software Version = BTS1
V111222\r\n      Current Software Status = Normal\r\n      Current Hot Patch Version =
V111222\r\n\r\n(Number of results = 1)\r\n\r\nDetail
information\r\n-----\r\nApplication Type  Application Version
Software Version          Application Hot Patch Version  Software Hot Patch
Version\r\n\r\nneNodeB          LTEV200R007C00SPC110  BTS1 V111222  V111222
BTS1 V111222\r\n(Number of results = 1)\r\n\r\n\r\n\r\n---   END\r\n\r\n",
    "result": "Operation succeeded.",
    "retCode": 0,
    "serialId": -1
  },
  {
    "name": "pTest",
    "report": "+++   LTE           2021-03-02 19:38:47\r\nO&M
#2440\r\n%/*1879048219*/LST VER:;%\r\nRETCODE = 0  Operation
succeeded.\r\n\r\nResult of current software
query\r\n-----\r\n      Current Software Version = BTS1
V111222\r\n      Current Software Status = Normal\r\n      Current Hot Patch Version =
V111222\r\n\r\n(Number of results = 1)\r\n\r\nDetail
information\r\n-----\r\nApplication Type  Application Version
Software Version          Application Hot Patch Version  Software Hot Patch
Version\r\n\r\nneNodeB          LTEV200R007C00SPC110  BTS1 V111222  V111222
BTS1 V111222\r\n(Number of results = 1)\r\n\r\n\r\n\r\n---   END\r\n\r\n",
    "result": "Operation succeeded.",
    "retCode": 0,
    "serialId": -1
  }
],
"retCode": "90000",
"retMessage": "Execution succeeded."
}
```

Operation Severity

Minor

5.2.1.1 Format of MML NE Response Messages

The response message body of an NE can be described in a horizontal list or a vertical list.

Horizontal List

The horizontal list is widely used and applicable to the display of multiple returned records that are not separated by blank lines. For example, it is used to display multiple returned results in the following scenarios: querying all the subracks of a module, querying the status of all the boards in a subrack, collecting entity traffic statistics, or querying the test results.

The format of a horizontal list is defined as follows:

```
<Attribute name 1'A0'Lw1><2SP><Attribute name 2'A0'Lw2><2SP>.....<2SP><Attribute  
name n'A0'Lwn>\r\n
```

```
\r\n
```

```
<Attribute 1 Value 1'A0'Lw1><2SP><Attribute 2 Value  
1'A0'Lw2><2SP>.....<2SP><Attribute n Value 1'A0'Lwn>\r\n
```

```
<Attribute 1 Value 2'A0'Lw1><2SP><Attribute 2 Value  
2'A0'Lw2><2SP>.....<2SP><Attribute n Value 2'A0'Lwn>\r\n
```

```
... ..
```

```
<Attribute 1 Value m'A0'Lw1><2SP><Attribute 2 Value  
m'A0'Lw2><2SP>.....<2SP><Attribute n Value m'A0'Lwn>\r\n
```

Note:

1. Lwn indicates the width of the attribute in column *n*. The maximum width of the attribute in each column is determined based on the report requirements. The minimum width is six characters.
2. The first line displays the attribute name of the output, which is separated from the following data area by a blank line.
3. The attribute of each row is left-aligned.
4. The attribute name and value cannot contain two or more consecutive spaces.
5. In the horizontal list, if an attribute of a record does not have the corresponding attribute value, **NULL** is used to indicate the attribute value.

Vertical List

The vertical list is applicable to the display of one or more returned records. Multiple returned records are separated by blank lines. For example, it is used to display returned results in the following scenarios: querying users or querying a test result.

Rule description:

1. In a vertical list, if each attribute has only one value, each row displays one attribute. The attribute name is on the left of the equal sign, and the value is on the right of the equal sign. The content on the left of the equal mark is right-aligned, and the content on the right of the equal mark is left-aligned. The format is as follows:

```
<0SP><Attribute name 1><2SP>=<2SP><Value 1>\r\n  
<0SP><Attribute name 2><2SP>=<2SP><Value 2>\r\n  
<0SP><Attribute name 3><2SP>=<2SP><Value 3>\r\n  
<0SP><Attribute name 4><2SP>=<2SP><Value 4>\r\n
```
2. In the vertical list, if an attribute does not have the corresponding attribute value, **NULL** is used to indicate the attribute value. The format is as follows:

<0SP><Attribute name><2SP>=<2SP><NULL>\r\n

3. In the vertical list, if an attribute has more than one value, each value is displayed in a line and the first value is used for left alignment. The format is as follows:

<0SP><Attribute name 1><2SP>=<2SP><Value 1>\r\n

<0SP><Attribute name 2><2SP>=<2SP><Value 2.1>\r\n

=<2SP><Value 2.2>\r\n

=<2SP><Value 2.3>\r\n

... ..

=<2SP><Value 2.x>\r\n

<0SP><Attribute name 3><2SP>=<2SP><Value 3>\r\n

<0SP><Attribute name 4><2SP>=<2SP><Value 4>\r\n

Note: < Value 2.x> indicates the xth value of the second attribute.

Message structure diagram

Figure 5-1 Horizontal list

```

+++-(1)      NE-(2)      2006-10-08-(3)  10:45:27-(4)
Q&M-(5)  #31420-(6)
%%/*9129*/ DSP BAMMODULE:;%(7)
RETCODE = 0-(8)      Execution succeeded.-(9)

Active BAM modules state-(10)
-----
Service Name-(11) State      Startup Type
Sntp Server-(12) Started    Automatic
Sntp Client      Started    Automatic-(14)
FTP Server       Stopped    Manual
DynMonitor       Started    Automatic
(Number of results = 4)-(15)

Standby BAM modules state
-----
Service Name      State      Startup Type
Sntp Server       Stopped    Disabled
Sntp Client       Started    Automatic
FTP Server        Started    Automatic
DynMonitor        Started    Automatic
(Number of results = 4)

---      END-(17)
  
```

Figure 5-2 Vertical list

```
+++ (1) NE (2) 2008-01-08 (3) 09:36:31 (4)
O&M (5) #7134 (6)
%%/*5398*/LST CELL: (7)
RETCODE = 0 (8) Execution succeeded (9)

List cell basic information (10)
-----
(11) Cell ID = 0 (12)
      Cell name = CELL1 (14)
      Local cell ID = 0
      DL primary scrambling code = 0 (13)

      Cell ID = 1
      Cell name = CELL2
      Local cell ID = 1
      DL primary scrambling code = 128
(Number of results = 2) (15)

--- END (17)
```

Note:

- 1. Start identifier
- 2. Source identifier (device name)
- 3. Message creation date
- 4. Creation time
- 5. Service report flag
- 6. Message sequence number
- 7. MML commands
- 8. Return code
- 9. Returned message
- 10. Title of the message entry
- 11. Attribute name
- 12. Attribute value
- 13. Message entry
- 14. Record in the entry
- 15. Result summary
- 16. Overall entry content
- 17. End flag

5.2.2 Issuing an MML Batch Script Task

5.2.2.1 Creating a Task

Function

This function allows you to create a script task based on the uploaded MML script file and the specified task name.

Precautions

- The task name must be a string of 1 to 64 characters and be unique, and cannot contain any character in `~!@#\$\$%^&\*(){}[]\|'.,?<>"`.
- The MML script file should be only in TXT format. The file content should contain the commands and NE information.
- The name of an MML script file cannot contain any character in `^<>\*?|'":'&`.
- The maximum size of the request body is 2 MB. Therefore, the size of the MML script file cannot exceed 2 MB.
- The script encryption password needs to be specified if the MML script contains the password field. Otherwise, an error is reported.
- New tasks cannot be created after the number of tasks reaches the maximum. The maximum number of tasks depends on the number of MML scripts on the **Task Management** page.

Call Method

POST

URI

/api/rest/mmlManagement/v1/tasks

Request Parameters

Table 5-11 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	STRING	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory
Content-Type	Request header	STRING	Type of the request message body. The value of this parameter is multipart/form-data .	Mandatory

Table 5-12 List of form parameters

Parameter	Mandatory	Type	Value Range	Default Value	Description
TaskRequest	Yes	REFERENCE	For details, see Table 5-13.		Parameter information for creating a task

Table 5-13 Parameters in the TaskRequest MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
file	Yes	STRING	-		MML script file.
taskName	No	STRING	-	-	Name of an MML script task. If the customer sends the task name, the task name is used when the script task is created. If the customer does not send the task name, the background automatically generates a task name.
secretKey	No	STRING	-	-	If the script file contains a password, this parameter is used to encrypt the file content.

Scripts Description

The format of an MML script is as follows:

*MML command:Parameters; {NE 1,NE 2,...} /*Comment*/*

1. One line in the MML script contains only one MML command.
2. MML command: parameter; is the command field, which cannot be empty. This field cannot contain the following characters: semicolon (;), two or more consecutive percent signs (%), two or more consecutive spaces, start characters of MML messages (+++), or end characters of MML messages (--- END).
3. {NE 1,NE 2, and ...} indicates the target NEs and can be empty for non-CloudEdge NEs. If {} is used, {} must be filled with NE names. You can enter multiple NE names separated by commas (,) in single-byte character (SBC) case. The NE name cannot contain the following characters in the SBC case: comma (,), colon (:), semicolon (;), open brace {}, and close brace (). For CloudEdge NEs, {NE 1,NE 2, and ...} indicates the target VNFC and must contain one VNFC name.
4. /*comment*/ is the comment field and can be omitted. This field cannot contain the following characters: two or more consecutive percent signs (%), two or more consecutive spaces, start characters of MML messages (+++), or end characters of MML messages (--- END).

The following provides an example of a correct MML script:

LST VER;; {HSS_1,HSS_2}

DSP PATCH:OT=BOARDTYPE,BT=UMPT; {HSS_1}

Sample Request

```
POST /api/rest/mmlManagement/v1/tasks HTTP/1.1
X-Auth-Token:
x-c9dilhbutj7u9h46hg5fjzlf2p2lio3svuti6rfvdf1dtdlidcqkipiqdi5etiukde7u47s5dcbzo9g6
852ofujx5g1fapanhdemg6ftl1ftilvuti48jvoapglipc2l
Host: *.*.*.*:31127
Content-Type: multipart/form-data;boundary=-----FormBoundaryShouldDifferAtRuntime
-----FormBoundaryShouldDifferAtRuntime
Content-Disposition: form-data; name="file"; filename="mml.txt"; taskName="mmlTask"
Content-Type: text/xml
[message-part-body; type: text/xml, size: 1924 bytes]
-----FormBoundaryShouldDifferAtRuntime--
```

Response Parameters

If the returned status code is **200**, the operation is successful.

Table 5-14 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
TaskResponse	No	REFERENCE	For details, see Table 5-15.		Task response parameter.

Table 5-15 Parameters in the TaskResponse MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
taskId	No	STRING	-	-	Unique task ID. This parameter is required for subsequent task operations.
retCode	No	STRING	-	-	Error code.
retMessage	No	STRING	-	-	Error message.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Date: Wed, 21 Nov 2018 10:00:00 GMT
Server: example-server
Content-Type: application/json
{
  "taskId": "123"
}
```

Operation Severity

Minor

5.2.2.2 Querying Task Status Based on the Task ID

Function

This function allows you to query the task status based on the returned task ID after an MML task is created successfully.

Precautions

- To query the task status, you need to use the task ID returned when the task is created.

Call Method

GET

URI

/api/rest/mmlManagement/v1/tasks/{taskId}/status

Request Parameters

Table 5-16 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	STRING	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory

Table 5-17 Path parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
taskId	Yes	STRING	-	-	Task ID.

Sample Request

```
GET /api/rest/mmlManagement/v1/tasks/123/status HTTP/1.1
X-Auth-Token:
x-jupj3v47dfhf4bpjnyermo2k9jpftdmmfveqgbmppf2kjwtgaikfwo5rskbjulisbbwk89ghcqovwruar
ftlefvanfy5ctes9hehi2mfyoa5grzmk9fldhcft07g52rka
Host: *.*.*.*:31127
```

Response Parameters

If the returned status code is **200**, the operation is successful.

Table 5-18 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
StatusResponse	No	REFERENCE	For details, see Table 5-19.		Task status information.

Table 5-19 Parameters in the StatusResponse MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
progress	No	INTEGER	0 to 100	-	Task progress.
currentState	No	INTEGER	-	-	Task status. 0: Idle 1: Running 2: Suspend 3: Complete
retCode	No	STRING	-	-	Error code.
retMessage	No	STRING	-	-	Error information.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Content-Type:application/json
{
  "progress": 100,
  "currentState": 3
}
```

Operation Severity

Minor

5.2.2.3 Downloading the Task Execution Result Based on the Task ID

Function

This function allows you to download the result file of a task when the task is complete.

Precautions

- The ID returned when the task is created needs to be used.
- If the task does not exist or is not complete, an error is reported.
- The maximum size of an MML result file is 500 MB. If the size of the result file is less than 100 MB, the system returns a .txt file. If the size of the result file is greater than 100 MB and less than 500 MB, the system compresses the file in ZIP format and returns it.
- It is recommended that a maximum of five file download requests be executed at the same time.

Call Method

GET

URI

/api/rest/mmlManagement/v1/tasks/{taskId}/result

Request Parameters

Table 5-20 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
-----------	----------	------	-------------	--------------------

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	STRING	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory

Table 5-21 Path parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
taskId	Yes	STRING	-	-	Task ID.

Sample Request

```
GET /api/rest/mmlManagement/v1/tasks/123/result HTTP/1.1
X-Auth-Token:
x-jupj3v47dfhf4bpjnyermo2k9jpftdmmfveqgbmppf2kjpgaikfwo5rskbjulisbbwk89ghcqovwruar
ftlefvanf5ctes9hehi2mfyoa5grzmk9fldhcft07g52rka
Host: *.*.*.*:31127
```

Response Parameters

If the returned status code is **200**, the operation is successful.

Table 5-22 Parameter list of successful transactions

Parameter	Mandatory	Type	Value Range	Default Value	Description
file	No	Streaming Output	-	-	File stream corresponding to the task execution result, which is returned when the transaction is successful.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Table 5-23 Parameter list of failed transactions

Parameter	Mandatory	Type	Value Range	Default Value	Description
retCode	No	STRING	-	-	Error code.
retMessage	No	STRING	-	-	Error information.

Result File Format

Figure 5-3 Content of a task execution result file

```

Script Task : test
=====Summary=====
Succeeded Number : 1 (1)
Failed Number : 0 (2)
=====

=====Failed MML Command===== (3)
=====

=====Succeeded MML Command===== (4)
MML Command-----LST VER:; (5)
NE : by (6)
(7) Report : +++ by 2021-03-02 19:38:47
O&M #2440
%%/*1879048219*/LST VER:;%
RETCODE = 0 Operation succeeded.

Result of current software query
-----
Current Software Version = BTS1 V111222
Current Software Status = Normal
Current Hot Patch Version = V111222

(Number of results = 1)

Detail information
-----
Application Type Application Version Software
eNodeB V111222 BTS1 V111222 V111222 BTS1 V11.
(Number of results = 1)

--- END

=====

```

Note:

1. Number of MML commands that are successfully executed
2. Number of MML commands that fail to be executed
3. Details about failed MML commands
4. Details about succeeded MML commands
5. MML command
6. NE name

7. NE response message. For details about the message format, see 5.2.1.1 Format of MML NE Response Messages.

Example of a Downloaded Result File

```
Script Task : test
=====Summary=====
Succeeded Number : 1
Failed Number : 0
=====

=====Failed MML Command=====
=====

=====Succeeded MML Command=====
MML Command-----LST VER:;
NE : by
Report : +++ by 2021-03-02 19:38:47
O&M #2440
%%/*1879048219*/LST VER:;%
RETCODE = 0 Operation succeeded.

Result of current software query
-----
Current Software Version = BTS1 V111222
Current Software Status = Normal
Current Hot Patch Version = V111222

(Number of results = 1)

Detail information
-----
Application Type Application Version Software Version Application Hot Patch
Version Software Hot Patch Version

eNodeB V111222 BTS1 V111222 V111222 BTS1 V111222
(Number of results = 1)

--- END

=====
```

Operation Severity

Minor

5.2.2.4 Deleting a Task Based on the Task ID

Function

This function allows you to delete the task after the task result is downloaded to avoid the failure to create a new task due to too many tasks.

Precautions

- The ID returned when the task is created needs to be used.
- A running task cannot be deleted.
- After a task is executed and the task result is downloaded, this API is called to delete the task to reduce system resource usage. If the task is not deleted immediately, MAE automatically deletes the task that has been created for more than two days.

Call Method

DELETE

URI

/api/rest/mmlManagement/v1/tasks/{taskId}

Request Parameters

Table 5-24 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	STRING	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory

Table 5-25 path parameters

Parameter	Mandatory	Type	Value Range	Default Value	Description
taskId	Yes	STRING	-	-	Task ID.

Sample Request

```
DELETE /api/rest/mmlManagement/v1/tasks/123 HTTP/1.1
X-Auth-Token:
x-jupj3v47dfhf4bpjnyermo2k9jpftdmmfveqgbmppf2kjwgaikfwo5rskbjulisbbwk89ghcqovwruar
ftlefvanfy5ctes9hehi2mfyoa5grzmk9fldhcf07g52rka
Host: *.*.*.*:31127
```

Response Parameters

If the returned status code is **200**, the operation is successful.

Table 5-26 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
retCode	No	STRING	-	-	Error code.
retMessage	No	STRING	-	-	Error information.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Content-Type:application/json
{
}
```

Operation Severity

Minor

5.3 List of Alarm APIs for FM

5.3.1 Querying Alarms

Function

This API is used to query alarms on the OSS, including current alarms, historical alarms, and alarm logs.

Precautions

- You are advised to use exact query with filter criteria (for example, specifying **AlarmId**) and avoid network-wide query without filter criteria.
- In scenarios where network-wide alarm data is queried, it is recommended that the query be performed only once a day or during off-peak hours of the MAE system.
- A maximum of five iterative queries can be performed for current alarms. A maximum of two iterative queries can be performed for historical alarms and alarm logs.

Call Method

GET

URI

/api/rest/faultSupervisionManagement/v1/alarms

Table 5-27 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	String	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory
Content-Type	Request header	String	Type of the request message body. The value of this parameter is application/json .	Mandatory

Table 5-28 Query parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
dataType	Yes	String	CURRENT/HISTORY/ALARM_LOG	-	Type of the alarm to be queried. The value is case insensitive. CURRENT : current alarms HISTORY : historical alarms ALARM_LOG : alarm logs (current and historical alarms)
alarmAck State	No	String	For details, see table 3.	Empty	Alarm acknowledgment status. The value is case insensitive.
baseObjectInstance	No	String	-	Empty	NE name or OSS. The corresponding field on the GUI is Alarm Source . When the NE name is specified, only one NE can be queried at a time.

Parameter	Mandatory	Type	Value Range	Default Value	Description
					If the OSS is specified, alarms reported by the OSS are queried.
filter	No	String	For details, see table 4.	Empty	Filter. For details about the fields, see table 4. Use URLEncode to encode the filter before transferring it. A maximum of three filter dimensions are supported.
specialAlarm	No	Boolean	true/false	false	Whether engineering alarms are supported.
limit	No	Integer	1-1000	500	Number of alarms displayed on a page. The actual number of reported alarms may be different from the value of limit .
marker	No	String	-	-	Iteration factor, which indicates whether there is a subsequent pagination flag. If this parameter is not empty, you need to enter the marker in the previous query result during the next iterative query

Remarks:

1. If neither **alarmAckState**, **baseObjectInstance**, nor **filter** is specified, all alarms are queried by data type by default.

Table 5-29 Enumeration list of alarmAckState character strings

Enumerated Value	Description
ALL_ALARMS	All alarms.
ALL_ACTIVE_ALARMS	All uncleared alarms.
ALL_ACTIVE_AND_ACKNOWLEDGED_ALARMS	All uncleared and acknowledged alarms.
ALL_ACTIVE_AND_UNACKNOWLEDGED_ALARMS	All uncleared and unacknowledged alarms.
ALL_CLEARED_AND_UNACKNOWLEDGED_ALARMS	All cleared and unacknowledged alarms.

Enumerated Value	Description
EDGED_ALARMS	
ALL_UNACKNOWLEDGED_ALARMS	All unacknowledged alarms.

Table 5-30 Field description of the filter character string

Field Name	Enumerated Value	Description
field	• alarmId: alarm ID (string) • alarmRaisedTime: UTC time when an alarm occurs (long) • alarmClearedTime: UTC time when an alarm is cleared (long) • ackTime: UTC time when an alarm is acknowledged (long) • csn: alarm log serial number (integer) • productName: NE type (string) • perceivedSeverity: alarm severity (integer)	Field condition for filtering. The value is case insensitive. • The character string supports the in, not in, =, and != operators: alarmId/productName. • The integer type supports the <, <=, >, >=, between, =, !=, in, and not in operators: alarmRaisedTime/alarmClearedTime/ackTime/csn/perceivedSeverity.
operator	• = (equal to) • < (less than) • <= (less than or equal to) • > (greater than) • >= (greater than or equal to) • != (not equal to) • in (belongs to a set) • not in (does not belong to a set) • between (between two digits)	Condition operator. If the operator is missing, an error is returned. The value is case insensitive.
values	-	Condition value. The following operators support a single value: =, <, <=, >, >=, and !=. The following operator supports double values: between. The following operators support multiple values: in and not

Field Name	Enumerated Value	Description
		in. A maximum of five value sets are supported.

Example of the filter Field

Filter data based on **csn** and **perceivedSeverity**. The relationship between multiple filter criteria is AND.

The JSON character string is as follows:

```
[
  {
    "field": "csn",
    "operator": "in",
    "values": ["19142", "19021"]
  },
  {
    "field": "perceivedSeverity",
    "operator": "in",
    "values": [1, 2]
  }
]
```

Encode the JSON character string using URLEncode and send the encoded character string to the filter.

filter=%5B%7B%22field%22%3A%22csn%22%2C%22operator%22%3A%22in%22%2C%22values%22%3A%5B%2219142%22%2C%2219021%22%5D%7D%2C%7B%22field%22%3A%22perceivedSeverity%22%2C%22operator%22%3A%22in%22%2C%22values%22%3A%5B1%2C2%2C3%5D%7D%5D

Sample Request

HTTP example:

```
HTTP/1.1
GET
/api/rest/faultSupervisionManagement/v1/alarms?dataType=CURRENT&limit=100
Host: serverIP:port
Content-Type: application/json
Accept: application/json
Accept-Language: en-US
X-AUTH-TOKEN:
CA48D152F6B19D84:637C38259E6974E17788348128A430FEE150E874752CE754B6BF855281219925
```

Response Parameters

If the returned status code is **200**, the alarm query is successful.

Table 5-31 Response parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
alarmInformationList	Yes	List of AlarmInformation	For details, see table 6.	-	Alarm information set to be queried.
status	Yes	String	OperationSucceeded/OperationFailed	-	Operation result.
marker	Yes	String	-	-	Iteration factor for the next query.
retCode	No	String	-	-	Return code.
retMessage	No	String	-	-	Returned message.

Table 5-32 Alarm information parameters

Parameter	Mandatory	Type	Description
alarmId	Yes	String	Alarm ID.
objectInstance	Yes	String	Alarm object instance, corresponding to Location Information on the GUI.
notificationType	Yes	String	Notification type. notifyNewAlarm : uncleared and unacknowledged alarms notifyChangedAlarm : acknowledged but uncleared alarm notifyClearedAlarm : cleared but unacknowledged alarm
alarmRaisedTime	Yes	String	Alarm generation time.
alarmClearedTime	Yes	String	Alarm clearance time, corresponding to Cleared On on the GUI.
alarmType	Yes	String	Alarm type, corresponding to

Parameter	Mandatory	Type	Description
			Type on the GUI. 1-power system, 2-Environment system 3-Signaling system, 4-Trunk system 5-Hardware system, 6-Software system 7-Running system, 8-Communication system 9-Qos, 10-Processing error 11-Internal, 12-Integrity violation 13-Operational violation, 14-Physical violation 15-Security service or mechanism violation 16-Time domain violation
probableCause	Yes	String	Possible cause of the alarm.
perceivedSeverity	Yes	String	Alarm severity, corresponding to Severity on the GUI. 1-Critical, 2-Major, 3-Minor, or 4-Warning
additionalInformation	No	String	Additional information in an event report, corresponding to Additional Information on the GUI.
ackTime	Yes	String	Alarm acknowledgment time, corresponding to Acknowledged On on the GUI.
ackUserId	Yes	String	ID of the user who acknowledges an alarm, corresponding to Acknowledged By on the GUI.
ackState	Yes	String	Acknowledgment status, corresponding to Acknowledgment Status on the GUI. 0-Unacknowledged or 1-Acknowledged
clearUserId	No	String	ID of the user who clears the alarm, which corresponds to

Parameter	Mandatory	Type	Description
			Cleared By on the GUI.
cleared	No	String	Clearance status, corresponding to Clearance Status on the GUI. 0-Uncleared or 1-Cleared
meName	No	String	NE name, corresponding to Alarm Source on the GUI.
productName	No	String	NE type, corresponding to NE Type on the GUI.
alarmName	No	String	Alarm name, corresponding to Name on the GUI.
nativeMoName	No	String	MO name, corresponding to MO Name on the GUI.
csn	No	String	Alarm serial number, corresponding to Log Serial Number on the GUI.
subCsn	No	String	Device alarm serial number, corresponding to Equipment Alarm Serial Number on the GUI.
specialAlarmStatus	No	String	Maintenance status, corresponding to Maintenance Status on the GUI.
comments	Yes	String	Remarks, corresponding to Comment on the GUI.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **429**, there are too many requests. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Date: Mon, 24 Dec 2018 10:00:00 GMT
```

```
Server: example-server
Content-Type: application/json
{
  "alarmInformationList": [
    {
      "alarmId": "4",
      "objectInstance": "Server name=OSSSVR01, SvcAgent=irppengine-3-1, SvcName=MAE/MAEAccessIRPPMEngine/mae_access_global, SiteName=Local Site",
      "notificationType": "notifyNewAlarm",
      "alarmRaisedTime": "1621859460000",
      "alarmClearedTime": "0",
      "alarmType": "10",
      "probableCause": "",
      "perceivedSeverity": "2",
      "additionalInformation": "IP address=172.28.129.6, Product alias=MAE",
      "ackTime": "0",
      "ackUserId": "",
      "ackState": "0",
      "clearUserId": "",
      "cleared": "0",
      "meName": "OSS",
      "productName": "OSS",
      "alarmName": "The OSS Service Is Terminated Abnormally",
      "nativeMoName": "OSS",
      "csn": "21559",
      "subCsn": "0",
      "specialAlarmStatus": "0",
      "comments": ""
    }
  ],
  "status": "OperationSucceeded",
  "marker": "MjAwJjg2MjI0OTMxNSM5ZDAzYjVhMi0yODQwLTQxZjktYjNjNC0zNjZiZjRiYjJlZjY=",
  "retCode": "90000",
  "retMessage": "Query succeeded."
}
```

Operation Severity

Minor

5.4 List of Performance APIs for PM

5.4.1 Creating a Performance Data Query Task

Function

This API is used to obtain performance data based on query conditions. If the data volume is small, performance data is directly returned through this API. If the data volume is large, task information is returned and performance data needs to be obtained asynchronously according to 5.4.2 Obtaining Performance Data.

Precautions

- Before creating a query task, ensure that the NE has issued the measurement subscription of the corresponding counters and the query period is the same as the period for issuing the subscription.
- After the performance result query is complete, you need to immediately invoke the function described in section 5.4.3 "Deleting a Performance Result Query Task" to delete the performance result query task to reduce the system resource usage.
- The time range of the query task must be within 24 hours. The number of counters to be queried cannot exceed 150, and the number of function subsets to which the counters belong cannot exceed 10.
- A maximum of five concurrent requests are supported.
- You are advised to specify an NE name to perform an exact query. Do not use the NE type to perform a network-wide query. If NE types are used for a network-wide query, it is recommended that the query be performed only once a day. In addition, do not query data of neighboring cells on the entire network.

Call Method

POST

URI

/api/rest/performanceManagement/v1/measurementResults

Table 5-33 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	STRING	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory
Content-Type	Request header	STRING	Type of the request message body. The value of this parameter is application/json .	Mandatory

Request Parameters

Table 5-34 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
condition	Yes	REFERENCE	For details, see table 3.	-	-

Table 5-35 Parameters in the condition MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
timeFormat	No	STRING			Query time format. The default value is timeString . This parameter is optional Value range: timeString : time string, for example, 2021-06-16 00:00:00 timeStampValue : Unix timestamp, for example, 1623772800
startTime	Yes	STRING	-	-	Start time of a query task. (The duration of the query task must be within 24 hours.)
endTime	Yes	STRING	-	-	End time of a query task. (The duration of the query task must be within 24 hours.)
period	Yes	INTEGER	-	-	Query period (unit: minute).
counterIds	Yes	Array<INTEGER>	-	-	List of counters to be queried (The number of counters cannot exceed 150, the counters belong to the same object type, and the number of function subsets to which the counters belong cannot exceed 10.)
isQueryAllNe	Yes	INTEGER	-	-	The value 1 indicates that network-wide NEs are queried. The value 0 indicates that specified NEs in neNames are queried. This parameter must be used together with neTypeName .
neTypeName	Yes	STRING	-	-	Name of the NE type to be queried. If isQueryAllNe is set to 0 , this parameter can be left empty. If isQueryAllNe is set to a value other than 0 , this parameter must be set to the type of the NE to be queried, and only one NE type can be specified at a time. The supported NE types are as

Parameter	Mandatory	Type	Value Range	Default Value	Description
					follows: BSC6900 GSM, BSC6900 GU, BSC6900 UMTS, BSC6910 GSM, BSC6910 GU, BSC6910 UMTS, BTS3900, BTS5900, micro BTS3900, pico BTS3900, eNodeB, NodeB, BTS3900 5G, BTS5900 5G, micro BTS5900, pico BTS3900, and pico BTS5900
neNames	Yes	Array<STRING>	-	-	List of NE names to be queried. If isQueryAllNe is set to 1 , this parameter can be set to an empty array. Otherwise, you need to specify the list of NE names to be queried. The NEs in a request must belong to one NE type.

Sample Request

HTTP example:

```
POST /api/rest/performanceManagement/v1/measurementResults HTTP/1.1
Host: *.*.*.*:31127
Content-Type: application/json
Accept: application/json
Accept-Language: en-US
X-AUTH-TOKEN:
CA48D152F6B19D84:637C38259E6974E17788348128A430FEE150E874752CE754B6BF855281219925
{
  "timeFormat": "timeString",
  "startTime": "2021-05-24 00:00:00",
  "endTime": "2021-05-24 00:15:00",
  "period": 60,
  "counterIds": [
    1526749447,
    1526743671
  ],
  "isQueryAllNe": 0,
  "neTypeName": "eNodeB",
  "neNames": [
    "FmaZcc"
  ]
}
```

Response Parameters

Parameter	Mandatory	Type	Value Range	Default Value	Description
retCode	No	STRING	-	-	Return code.
retMessage	No	STRING	-	-	Returned message.
taskId	No	STRING	-	-	Task ID. If the task is created successfully, the task ID is returned. Otherwise, this parameter is not returned.

If the returned status code is **200**, the request is successful.

If the returned status code is **202**, the request is successful, but the query result is not ready. The result needs to be obtained in asynchronous mode.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **429**, the request frequency exceeds the flow control threshold.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

The task is created successfully, and the first batch of data is returned. For details about the data structure, see Table 5-38.

```
HTTP/1.1 200 OK
Mon, 24 May 2021 08:41:26 GMT
Server: example-server
Content-Type: application/json
{
  "counterIds": [
    1526749447,
    1526743671
  ],
  "marker": "null",
  "period": 60,
  "result": [
    {
```

```

        "counterValues": [
            "77",
            "71"
        ],
        "neFdn": "NE=257",
        "neName": "FmaZcc",
        "objectName": "eNodeB Function Name=31_15, Local Cell ID=3, Cell Name=CELL3,
eNodeB ID=158435, Cell FDD TDD indication=CELL_TDD",
        "startTime": "2021-05-24 00:00:00"
    },
    {
        "counterValues": [
            "77",
            "85"
        ],
        "neFdn": "NE=257",
        "neName": "FmaZcc",
        "objectName": "eNodeB Function Name=31_15, Local Cell ID=2, Cell Name=CELL2,
eNodeB ID=158435, Cell FDD TDD indication=CELL_TDD",
        "startTime": "2021-05-24 00:00:00"
    }
],
"retCode": "90000",
"retMessage": "Completed.",
"status": 2,
"taskId": "16",
"totalSize": 2
}

```

The task is created successfully, but the data is not returned immediately.

```

HTTP/1.1 202 OK
Content-Type:application/json
{
    "retCode": "90037",
    "retMessage": "The data is being collected.",
    "taskId": "1569825"
}

```

The task fails to be created.

```

HTTP/1.1 400 OK
Content-Type:application/json
{
    "retCode": "90052",
    "retMessage": "The Parameter period value is invalid."
}

```

Operation Severity

Minor

5.4.2 Obtaining Performance Data

Function

If a large amount of data is queried, this API needs to be used to obtain performance data. If the query task is complete, the performance data is returned. Otherwise, polling is required and the performance data can be queried only after the task is complete.

Precautions

- The marker can be left empty except for the first query. Otherwise, the marker returned by the current task in the last batch of performance result data needs to be used for each query. If the markers are inconsistent, the performance result data cannot be obtained.

Call Method

GET

URI

/api/rest/performanceManagement/v1/measurementResults/{taskId}

Table 5-36 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	String	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory
Content-Type	Request header	String	Type of the request message body. The value of this parameter is application/json .	Mandatory

Request Parameters

Table 5-37 Path input parameters

Parameter	Location	Mandatory	Type	Description
taskId	Path parameter	Yes	STRING	Unique ID of a query task.
limit	Path parameter	No	INTEGER	Number of records queried at a time. The maximum number is 5000. The default value is 5000 .

Parameter	Location	Mandatory	Type	Description
marker	Path parameter	No	STRING	Mark to iterate to the next page. It can be empty for the first query. If more data needs to be iterated on the next page, the marker tag may be set in the first response. Then, the NMS should use the tag for the next calling to obtain the next page. If there is no more data, the last page (response) sets marker tag to null .

Sample Request

First query:

```
GET /api/rest/performanceManagement/v1/measurementResults/23456?limit=5000
HTTP/1.1
X-Auth-Token:
x-rxvuk4juns6nv2lg9rz7s44g4oamkmoo5lj0bipo5fzkhc5f9cnz46ljenmrlijthgdf5hph3vlgghg6pffw49hdob7w8bo54a21899ho7c5emg6vu9jtdg6050bph
Host: *.*.*.*:31127
```

Use the marker to query more batches of result data:

```
GET
/api/rest/performanceManagement/v1/measurementResults/234571?limit=5000&marker=8acea8ca-0dcb-11ea-8000-0050568aff80
HTTP/1.1
X-Auth-Token:
x-rxvuk4juns6nv2lg9rz7s44g4oamkmoo5lj0bipo5fzkhc5f9cnz46ljenmrlijthgdf5hph3vlgghg6pffw49hdob7w8bo54a21899ho7c5emg6vu9jtdg6050bph
Host: *.*.*.*:31127
```

Response Parameters

Table 5-38 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
body	Yes	ARRAY_REFERENCE	For details, see Table 5-39.	-	

Table 5-39 Body parameter list

Parameter	M a n d a t o r y	Type	Val ue Ran ge	Default Value	Description
status	N o	INTEGER	-	-	Task status. The value 2 indicates success.
retCode	N o	STRING			Return code.
retMessage	N o	STRING			Returned message.
taskId	N o	STRING			Task ID.
marker	N o	STRING			Result ID. This parameter is required when you continue to obtain data. If the value is null , the data has been obtained.
period	N o	INTEGER	-	-	Period. If the task status is 2 , this parameter is displayed.
totalSize	N o	INTEGER	-	-	Total number of obtained records. If the task status is 2 , this parameter is displayed.
counterIds	N o	Array<INTEGER>	-	-	List of counters to be queried. If the task status is 2 , this parameter is displayed.
result	N o	ARRAY_REFERENCE	For details, see Table 5-40.	-	List of performance results obtained after query. If the task status is 2 , this parameter is displayed.

Table 5-40 Result parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
startTime	Yes	STRING	-	-	Start time.
neName	Yes	STRING	-	-	NE name.
neFdn	Yes	STRING	-	-	NE FDN.
objectName	Yes	STRING	-	-	Object name.
counterValues	Yes	Array<STRING>	-	-	Result corresponding to the counter ID list. The invalid value is NIL .

If the returned status code is **200**, the request is successful.

If the returned status code is **202**, the request is successful, but the query result is not ready. In this case, you need to try again.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response 200

```
HTTP/1.1 200 OK
Date: Sat, 22 May 2021 09:25:37 GMT
Server: example-server
Content-Type: application/json
{
  "counterIds": [
    1526728519,
    1526728520,
    1526728518,
    1526728514,
    1526728515,
    1526730594,
    1526730595,
    1526730592,
    1526730593,
    1526730596,
    1526730597,
```

```

        1526749439,
        1526737790
    ],
    "marker": "aaf85822-badf-11eb-8000-fa163eb450dd",
    "period": 15,
    "result": [
        {
            "counterValues": [
                "108",
                "69",
                "313",
                "338",
                "145",
                "983",
                "402",
                "657",
                "818",
                "472",
                "762",
                "NIL",
                "338"
            ],
            "neFdn": "NE=24327",
            "neName": "lx test",
            "objectName": "eNodeB Function Name=FM nsjiahao, Local Cell ID=2, Cell
Name=FM nsjiahao 2, eNodeB ID=829, Cell FDD TDD indication=CELL FDD",
            "startTime": "2021-05-19 16:00:00"
        },
        .....
        {
            "counterValues": [
                "330",
                "220",
                "969",
                "595",
                "339",
                "248",
                "299",
                "892",
                "588",
                "83",
                "341",
                "NIL",
                "598"
            ],
            "neFdn": "NE=13234",
            "neName": "LTE 100 005 176",
            "objectName": "eNodeB Function Name=FM nsjiahao, Local Cell ID=0, Cell
Name=FM nsjiahao, eNodeB ID=829, Cell FDD TDD indication=CELL FDD",
            "startTime": "2021-05-19 00:00:00"
        }
    ],
    "retCode": "90000",
    "retMessage": "Completed.",
    "status": 2,

```

```
"taskId": "3",  
"totalSize": 5000  
}
```

Sample Response 202

```
HTTP/1.1 202 OK  
Content-Type:application/json  
{  
  "retCode": "90037",  
  "retMessage": "The data is being collected.",  
  "taskId": "123"  
}
```

Operation Severity

Minor

5.4.3 Deleting a Performance Result Query Task

Function

This API is used to delete a task whose performance results have been obtained based on the task ID to reduce the system resource usage.

Precautions

- If this API is not called to delete a task, MAE automatically deletes the task 12 hours after the task is created. If you want to continue to obtain data, you need to create a query task again.

Call Method

DELETE

URI

/api/rest/performanceManagement/v1/measurementResults/{taskId}

Table 5-41 Header parameter list

Parameter	Location	Mandatory/Optional	Type	Description
X-Auth-Token	Request header	Mandatory	STRING	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.

Parameter	Location	Mandatory/Optional	Type	Description
Content-Type	Request header	Mandatory	STRING	Type of the request message body. The value of this parameter is application/json .

Request Parameters

Parameter	Location	Mandatory	Type	Value Range	Default Value	Description
taskId	Path parameter	Yes	STRING	-	-	ID of the task to be deleted.

Sample Request

```
DELETE /api/rest/performanceManagement/v1/measurementResults/0220000052 HTTP/1.1
X-Auth-Token:
x-rxvuk4juns6nv2lg9rz7s44g4oamkmoo5lj0bipo5fzukahc5f9cnz46ljenmrlijthgdf5hph3vlghg
g6pffw49hdob7w8bo54a21899ho7c5emg6vu9jtdg6050bph
Host: *.*.*.*:31127
```

Response Parameters

Parameter	Mandatory	Type	Value Range	Default Value	Description
retCode	No	STRING	-	-	Return code.
retMessage	No	STRING	-	-	Returned message.
taskId	No	STRING	-	-	Task ID.

If the returned status code is **200**, the request is successful.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

The task is deleted successfully.

```
HTTP/1.1 200 OK
Content-Type:application/json
{
  "retCode" : "90000",
  "retMessage" : "Completed.",
  "taskId" : "3"
}
```

The task fails to be deleted.

```
HTTP/1.1 404 OK
Content-Type:application/json
{
  "retCode" : "90048",
  "retMessage" : "Task id does not exist."
}
```

Operation Severity

Minor

5.5 API for Querying Topology Cell Information

Function

This API is used to obtain the list of cells in the current topology on the OSS.

Precautions

- In the request message, it is recommended that the number of specified FDNs be less than or equal to 500. The timeout interval for requesting the bus is 5 minutes. If there are too many NEs, the request may time out.

Call Method

POST

URI

/api/rest/resourceManagement/v1/topocellsinfo

Request Parameters

Table 5-42 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
X-Auth-Token	Request header	STRING	Login session ID. The value of this parameter is the same as that of the accessSession parameter in the response message body for the API used for obtaining the token.	Mandatory
Content-Type	Request header	STRING	Type of the request message body. The value of this parameter is application/json .	Mandatory

Table 5-43 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
TopocellRequest	Yes	REFERENCE	For details, see Table 5-44.		Request message for querying topology cell information.

Table 5-44 Parameters in the TopocellRequest MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
fdns	Yes	ARRAY OF STRING	-	-	FDN list of a base station.

Sample Request

```
POST /api/rest/resourceManagement/v1/topocellsinfo
Host: serverIP:port
Content-Type: application/json; charset=UTF-8
X-Auth-Token:
x-jupj3v47dfhf4bpjnnyermo2k9jpftdmmfveqgbmppf2kjwtgaikfwo5rskbjulisbbwk89ghcqvruar
ftlefvanfy5ctes9hehi2mfyoa5grzmk9fldhcft07g52rka
Content-Length:...
{
  "fdns": [
    "NE=1201"
```

```
]
}
```

Response Parameters

If the returned status code is **200**, the operation is successful.

Table 5-45 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
results	No	ARRAY OF NodeCellInfo	For details, see Table 5-46.		Set of queried cell information.

Table 5-46 Parameters in the NodeCellInfoSeq MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
logicBTS Dn	Yes	STRING	-	-	Base station DN.
info	Yes	STRING	-	-	Base station information.
neDn	Yes	STRING	-	-	NE DN.
objectId	Yes	STRING	-	-	eNodeB ID.
cellInfos	Yes	ARRAY OF CellInfo	For details, see Table 5-47.		List of topology cell information.

Table 5-47 Parameters in the CellInfo MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
objDn	Yes	STRING	-	-	Cell DN.
cellId	Yes	STRING	-	-	Cell ID.
stateVals	Yes	ARRAY OF STRING	-	-	For details, see Table 5-48.
results	Yes	ARRAY OF STRING	-	-	Result character string array, which is used to display cell information and status information.

Parameter	Mandatory	Type	Value Range	Default Value	Description
					The following information is displayed in sequence: RAT, cell ID, cell name, administrative status, activation status, operation status, availability status, and prompt information.

Table 5-48 Enumeration list of the stateVals character string

Status Code	Enumerated Value	Description
0	ADMIN_UNLOCKED	Unlocked
1	ADMIN_LOCKED	Blocked
2	ACT_ACTIVE	Activated
3	ACT_DEACTIVE	Deactivated
4	ADMIN_UNKOWN	Unknown

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **500**, the server is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=UTF-8
Content-Length:...
{
  "results": [
    {
      "logicBTSDn": "NE=1201",
      "info": "",
      "neDn": "NE=1201",
      "objectId": 1003224404058,
      "cellInfos": [
        {
          "objDn": "NE=1201,eNodeBCell=0",
          "cellId": 0,
```

```
    "stateVals": [
      0,
      1,
    ],
    "results": [
      "LTE (FDD) ",
      "0",
      "dualMod",
      "Unlocked",
      "Inactive",
      "Enable",
      "Invalid",
      ""
    ]
  },
  {
    "objDn": "NE=1201,eNodeBCell=1",
    "cellId": 1,
    "stateVals": [
      0,
      2,
    ],
    "results": [
      "LTE (FDD) ",
      "1",
      "dualMod 1",
      "Unlocked",
      "Inactive",
      "Enable",
      "Invalid",
      ""
    ]
  }
]
}
```

Operation Severity

Minor

5.6 List of iSStar APIs

5.6.1 Creating iSStar Task

Function

This API is used to create a task based on the iSStar script uploaded according to 4.3 (Optional) Deploying iSStar Scripts.

Precautions

- Currently, the maximum number of concurrent iSStar tasks is 75, and the maximum number of tasks is 200.
- The tasks will be deleted 48 hours after the execution is complete.

Call Method

POST

URI

/api/rest/scriptsManagement/v1/tasks

Request Parameters

Table 5-49 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
RestCreateJobInputs	Yes	REFERENCE	For details, see Table 5-50.		Input parameters required for creating an iSStar5G report generation task.

Table 5-50 Parameters in the RestCreateJobInputs MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
entryScript	Yes	STRING	-	-	Name of the script to be deployed.
scriptPath	Yes	STRING	-	-	Relative path of the script.
startTime	No	STRING	-	-	Script execution time. By default, the script is executed immediately. Define the UTC time based on ISO8601. The format is as follows: <YYYY-MM-DD>T<HH:MM:SS>Z For example: 2019-11-12T01:19:29Z
scriptHashValue	Yes	STRING	-	-	Hash value of the script.
scriptParams	No	List			Parameter used for running

Parameter	Mandatory	Type	Value Range	Default Value	Description
					the script. The parameter is in the standard JSON format. Table 5-51 lists an example.

Table 5-51 Parameters in the scriptParam MO

Parameter	Mandatory	Type	Value Range	Default Value	Description
<i>data</i>	No	String	-	-	List of parameters used by iSStar applications, which is a standard JSON character string.

Sample Request

HTTP example:

```
POST /api/rest/scriptsManagement/v1/tasks HTTP/1.1
Host: *.*.*.*:31127
Content-Type: application/json
Accept: application/json
Accept-Language: en-US
X-AUTH-TOKEN:
CA48D152F6B19D84:637C38259E6974E17788348128A430FEE150E874752CE754B6BF855281219925
{
  "entryScript": "test.hsp3",
  "scriptPath": "demo",
  "startTime": "string",
  "scriptHashValue":
  "*****",
  "scriptParams": [
    {
      "data": "test"
    }
  ]
}
```

Response Parameters

Table 5-52 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
-----------	-----------	------	-------------	---------------	-------------

Parameter	Mandatory	Type	Value Range	Default Value	Description
data	No	STRING	For details, see Table 5-53.		Task list. Currently, the value is fixed to a single task.
retCode	No	STRING	-	-	Error code.
retMessage	No	STRING	-	-	Error information.

Table 5-53 Parameters in the jobs MO

Parameter	Type	Value Range	Default Value	Description
id	STRING	-	-	Task ID.
retCode	STRING	-	-	0 : The task is executed successfully. 1001 : The parameter for task creation is invalid. 1002 : The hash value for task creation is invalid. 1003 : The start time for task creation is invalid. 1004 : The file type of the task creation code is invalid. 1005 : The file for creating the task does not exist. 1010 : Other errors.
retMessage	STRING	-	-	Result message.
href	STRING	-	-	href for the task.
jobStatus	STRING	-	-	Idle : The task has been created but has not been executed (on a scheduled basis). Running : The task is running. Finished : The task is complete. NotExist : The task does not exist. Unknown : The status is abnormal
jobExecResult	STRING	-	-	Processing : The task is running. Failed : The task fails to be executed. Successful : The task is executed successfully.

Parameter	Type	Value Range	Default Value	Description
				NotExist: The task does not exist. Unknown: The running result is abnormal.
userName	STRING	-	-	Name of the user who creates tasks.

If the returned status code is **200**, the task creation is successful.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid. For details, see the response message body.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the request fails to be processed. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Date: Tue, 7 Jan 2020 10:00:00 GMT
Server: example-server
Content-Type: application/json
{
  "data": [
    {
      "jobStatus": "Running",
      "retMessage": "Process task successfully.",
      "id": "15103",
      "href": "/mnf/10.185.172.143/api/rest/scriptsManagement/v1/tasks/15103",
      "retCode": "0",
      "username": "northAPIUser"
    }
  ]
}
```

Operation Severity

Minor

5.6.2 Querying iSStar Tasks

Function

This API is used to obtain the iSStar task execution result based on the task ID.

Precautions

- If the size of the returned result is less than 10 MB, the result is returned directly. If the size is greater than 10 MB, you need to download the iSStar task result report by following the instructions provided in 5.6.4 Downloading the iSStar Task Result Report.

Call Method

GET

URI

/api/rest/scriptsManagement/v1/tasks/{jobId}

Table 5-54 Query parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
jobId	Yes	STRING	-	-	Task ID.

Request Parameters

None

Sample Request

HTTP example:

```
GET /api/rest/scriptsManagement/v1/tasks/{jobId} HTTP/1.1
Host: *.*.*.*:31127
Content-Type: application/json
Accept: application/json
Accept-Language: en-US
X-AUTH-TOKEN:
CA48D152F6B19D84:637C38259E6974E17788348128A430FEE150E874752CE754B6BF855281219925
```

Response Parameters

Table 5-55 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
-----------	-----------	------	-------------	---------------	-------------

Parameter	Mandatory	Type	Value Range	Default Value	Description
data	No	STRING	For details, see Table 5-56.		Task list. Currently, the value is fixed to a single task.
retCode	No	STRING	-	-	Error code.
retMessage	No	STRING	-	-	Error information.

Table 5-56 Parameters in the data MO

Parameter	Type	Value Range	Default Value	Description
id	STRING	-	-	Task ID.
retCode	STRING	-	-	0 : The task is complete. 2001 : The task does not exist. 2002 : Other errors.
href	STRING	-	-	href for the task.
jobStatus	STRING	-	-	Idle : The task has been created but has not been executed (on a scheduled basis). Running : The task is running. Finished : The task is complete. NotExist : The task does not exist. Unknown : The status is abnormal
retMessage	STRING	-	-	Result message.
jobExecResult	STRING	-	-	Processing : The task is running. Failed : The task fails to be executed. Successful : The task is executed successfully. NotExist : The task does not exist. Unknown : The running result is abnormal.
endTimeGmt	STRING	-	-	Task end time.
reportUri	STRING	-	-	Task execution result file (displayed after the task execution is complete).
progress	STRING	-	-	Task execution progress.

Parameter	Type	Value Range	Default Value	Description
userName	STRING	-	-	Name of the user who performs the query.
extInfo	STRING			Result information generated during script execution. If the size is less than 10 MB, the result information is displayed directly. If the size is greater than 10 MB, no result information is displayed.
extCode	STRING			Result of obtaining the script execution result. 0: The information is obtained successfully. 1: The file does not exist. 2: The file size exceeds 10 MB. 3: The information fails to be obtained.

If the returned status code is **200**, the query is successful.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid. For details, see the response message body.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the request fails to be processed. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Date: Tue, 7 Jan 2020 10:00:00 GMT
Server: example-server
Content-Type: application/json
{
  "data": [
    {
      "jobStatus": "Finished",
      "extCode": "1",
      "retMessage": "Process task successfully.",
      "endTimeGmt": "2021-05-20T09:48:56GMT+08:00",
      "reportUri":
"/mnf/10.185.172.143/api/rest/scriptsManagement/v1/files/15108/result.zip",
      "progress": "100%",
      "id": "15108",
```

```
        "href": "/mnf/10.185.172.143/api/rest/scriptsManagement/v1/tasks/15108",  
        "retCode": "0",  
        "jobExecResult": "Failed",  
        "username": "northAPIUser",  
        "extInfo": ""  
    }  
]  
}
```

Operation Severity

Minor

5.6.3 Deleting iSStar Tasks

Function

This API is used to delete a completed task based on the task ID to reduce system resource usage.

Precautions

- After a task is executed and the task result is downloaded, this API is called to delete the task to reduce system resource usage. If the task is not deleted immediately, MAE automatically deletes the task that has been completed for more than two days.

Call Method

DELETE

URI

/api/rest/scriptsManagement/v1/tasks/{jobId}

Table 5-57 Query parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
jobId	Yes	STRING	-	-	Task ID.

Request Parameters

None

Sample Request

HTTP example:

```
DELETE /api/rest/scriptsManagement/v1/tasks/{jobId} HTTP/1.1  
Host: *.*.*:31127  
Content-Type: application/json
```

```
Accept: application/json
Accept-Language: en-US
X-AUTH-TOKEN:
CA48D152F6B19D84:637C38259E6974E17788348128A430FEE150E874752CE754B6BF855281219925
```

Delete Data

Task execution result and task ID in the task list.

Response Parameters

Table 5-58 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
data	No	STRING	For details, see Table 5-59.		Task list. Currently, the value is fixed to a single task.
retCode	No	STRING	-	-	Error code.
retMessage	No	STRING	-	-	Error information.

Table 5-59 Parameters in the jobs MO

Parameter	Type	Value Range	Default Value	Description
id	STRING	-	-	Task ID.
href	STRING	-	-	href for the task.
retCode	STRING	-	-	0 : The task is complete. 3001 : The task does not exist. 3002 : The task is not complete. 3003 : Other exceptions occur.
retMessage	STRING	-	-	Result message.
username	STRING	-	-	Name of the user who performs the deletion.

If the returned status code is **200**, the deletion is successful.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid. For details, see the response message body.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the request fails to be processed. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Date: Tue, 7 Jan 2020 10:00:00 GMT
Server: example-server
Content-Type: application/json
{
  "data": [
    {
      "retMessage": "Process task successfully.",
      "id": "15108",
      "href": "/mnf/10.185.172.143/api/rest/scriptsManagement/v1/tasks/15108",
      "retCode": "0",
      "username": "northAPIUser"
    }
  ]
}
```

Operation Severity

Minor

5.6.4 Downloading the iSStar Task Result Report

Function

This API is used to download the result of a task that has been executed based on the task ID.

Precautions

- You can download the result report of a task only after the query task is complete.

Call Method

GET

URI

/mnf/{IP}/api/rest/scriptsManagement/v1/files/{jobId}/result.zip

Table 5-60 path parameters

Parameter	Mandatory	Type	Value Range	Default Value	Description
jobId	Yes	INTEGER	-	-	-
IP	Yes	STRING	-	-	IP address of the MAE master server.

Request Parameters

None

Sample Request

HTTP example:

```
GET /mnf/{IP}/api/rest/scriptsManagement/v1/files/{jobId}/result.zip HTTP/1.1
Host: *.*.*.*:31127
Content-Type: application/json
Accept: application/json
Accept-Language: en-US
X-AUTH-TOKEN:
CA48D152F6B19D84:637C38259E6974E17788348128A430FEE150E874752CE754B6BF855281219925
```

Response Parameters

If the returned status code is **200**, the download is successful.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid. For details, see the response message body.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **404**, the requested resource does not exist or the URI is incorrect. For details, see the response message body.

If the returned status code is **500**, the request fails to be processed. For details, see the response message body.

Operation Severity

Minor

5.7 List of System Information APIs

5.7.1 Testing MAE Northbound Connectivity

Function

This API uses the token information returned in 5.1.1 Login API to test the MAE northbound connectivity in real time.

Precautions

You have called the API in 5.1.1 Login API to obtain the token information.

Call Method

POST

URI

/api/rest/oss-info/v1/connection-status

Request Parameters

Table 5-61 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
accessToken	Request header	STRING	accessSession returned after a successful login.	Mandatory

Sample Request

```
POST /api/rest/oss-info/v1/connection-status HTTP/1.1
Host: serverIP:port
accessToken: 52661fbd-6b84-4fc2-aa1e-17879a5c6c9b
Content-Type: application/json; charset=UTF-8
Content-Length:...
{
}
```

Response Parameters

If the returned status code is **200**, the API is successfully called.

Table 5-62 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
-----------	-----------	------	-------------	---------------	-------------

Parameter	Mandatory	Type	Value Range	Default Value	Description
retCode	Yes	int	0	0	0-The connectivity is normal.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **429**, there are too many requests. For details, see the response message body.

If the returned status code is **500**, the service is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Content-Length:...
{
  "retCode": 0
}
```

Operation Severity

Minor

5.7.2 Querying the OMC Number of MAE

Function

This API uses the token information returned in 5.1.1 Login API to query the OMC number of MAE.

Precautions

You have called the API in 5.1.1 Login API to obtain the token information.

Call Method

GET

URI

/api/rest/oss-info/v1/omcid

Request Parameters

Table 5-63 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
accessToken	Request header	STRING	accessSession returned after a successful login.	Mandatory

Sample Request

```
POST /api/rest/oss-info/v1/omcid HTTP/1.1
Host: serverIP:port
accessToken: 52661fbd-6b84-4fc2-aa1e-17879a5c6c9b
Content-Type: application/json; charset=UTF-8
Content-Length:...
{
}
```

Response Parameters

If the returned status code is **200**, the API is successfully called.

Table 5-64 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
omcId	Yes	STRING			Returned OMC number of MAE. A null character is returned if no OMC number is set.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **429**, there are too many requests. For details, see the response message body.

If the returned status code is **500**, the service is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
```

```
Content-Type: application/json;charset=UTF-8
Content-Length:...
{
  "omcId": "X"
}
```

Operation Severity

Minor

5.7.3 Querying the Current MAE Version

Function

This API uses the token information returned in 5.1.1 Login API to query the current MAE version.

Precautions

You have called the API in 5.1.1 Login API to obtain the token information.

Call Method

GET

URI

/api/rest/oss-info/v1/mae-version

Request Parameters

Table 5-65 Header parameter list

Parameter	Location	Type	Description	Mandatory/Optional
accessToken	Request header	STRING	accessToken returned after a successful login.	Mandatory

Sample Request

```
POST /api/rest/oss-info/v1/mae-version HTTP/1.1
Host: serverIP:port
accessToken: 52661fbd-6b84-4fc2-aa1e-17879a5c6c9b
Content-Type: application/json; charset=UTF-8
Content-Length:...
{
}
```

Response Parameters

If the returned status code is **200**, the API is successfully called.

Table 5-66 Body parameter list

Parameter	Mandatory	Type	Value Range	Default Value	Description
version	Yes	STRING			Current MAE version number that is returned.

If the returned status code is **400**, the request is invalid. For details, see the response message body.

If the returned status code is **401**, the operation is unauthorized and accessSession is invalid.

If the returned status code is **403**, the requester does not have the access permission. For details, see the response message body.

If the returned status code is **429**, there are too many requests. For details, see the response message body.

If the returned status code is **500**, the service is abnormal. For details, see the response message body.

Sample Response

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Content-Length:...
{
  "version": "V100R021C10"
}
```

Operation Severity

Minor

6 Error Code List

Error Code	Description
90000	Export successful
90001	Parameter \$0 does not exist.
90003	Failed to register the job.
90005	Parameter \$0 can not be empty.
90011	Exception: \$0
90016	The \$0 format error
90032	The task name already exists.
90033	The taskName is invalid.
90034	Data does not exist.
90042	Completed, no data exported
90044	Marker is wrong
90048	The task id does not exist
90051	File is invalid
90052	Invalid value for parameter: \$0
90064	Job has exceed the max number \$0.
92000	Partially completed:\$0
99000	Invalid request:\$0
99010	Invalid credentials:\$0
99030	Access denied:\$0
99040	Url not found:\$0
99050	Method not allowed.
99998	Export failed.

